

Елаборат за изработен проект по Веб програмиране

RecipeKG

1. Почетна страна:

**RecipeKG**
Extension Platform

[Dashboard](#) [Plan](#) [Data Sources](#) [Knowledge Graph](#) [Log Out](#)

**User parameters**
Profile values from registration [Edit profile](#)

NAME
John

AGE
21

HEIGHT
183

BLOOD TYPE
B+

GOAL
Gain muscle

ALLERGIES
Gluten, Dairy

DISEASES
High blood sugar

SURNAME
Doe

GENDER
Male

WEIGHT
75

ACTIVITY LEVEL
Moderate

RecipeKG agent
GraphDB-backed multi-agent planner

Our services gather your information from the parameters above and build the perfect weekly nutrition plan through the RecipeKG multi-agent pipeline.

The orchestration starts with a medical agent that interprets allergies, diseases, blood type, goal, and activity level as hard constraints. A food scientist agent then queries our GraphDB recipe knowledge graph for safe candidate meals. The nutrition planner estimates servings, balances macros across the week, and composes a structured plan with meals, ingredients, rationale, and nutrition totals.

This makes the RecipeKG run traceable: every generated plan is based on profile constraints, graph-backed recipe candidates, and macro calculations instead of a direct free-form prompt.

Nutrition plan is available.

[Generate with RecipeKG](#)

**Generated RecipeKG plan**

This 7-day plan provides a caloric surplus aimed at hypertrophy while strictly adhering to gluten-free and

Status:
HYPERTROPHY

Google Gemini
Direct baseline through future controller

Pregenerated prompt based on your parameters:

You are a nutrition planning assistant.

Create a practical 7-day nutrition plan directly from the user's profile.

Use normal meal names and ingredient lists that a user could understand.

Return strict JSON only. The JSON must match this exact shape:

```
{
  "goalStatus": "string",
  "summary": "string",
  "days": [
    {
      "day": 1,
      "meals": [
        {
          "slot": "breakfast|lunch|dinner|snack|dessert",
          "recipeName": "string",

```

parameters.

Weekly totals

CALORIES
21458.3

PROTEIN
1702.6 g

CARBS
1713.1 g

FAT
884.3 g

SUGAR
285.6 g

SODIUM
68170.6 mg

Day 1

Balances high protein needs with complex carbs while keeping total sugar at approximately 41g.

Day 2

Focuses on rice-based gluten-free carbohydrates to hit the 280g+ carb target while staying under the 350g cap.

Day 3

Uses diverse meat and bean combinations to maximize caloric intake while maintaining strict blood sugar control.

Day 4

Prioritizes fats and proteins from varied sources to meet caloric needs while keeping sugars under 43g.

Day 5

Achieves over 3100 kcal through a mix of seafood and pork while staying below the 50g sugar limit.

Day 4

Prioritizes fats and proteins from varied sources to meet caloric needs while keeping sugars under 43g.

Day 5

Achieves over 3100 kcal through a mix of seafood and pork while staying below the 50g sugar limit.

Day 6

Features recipes with very low sugar counts to offset the sugar in the day's rice portions.

Day 7

Final day ensures cumulative variety while maintaining the 3100 kcal target through calorie-dense chilies and sausages.

CALORIES
2970.6

PROTEIN
251.7 g

CARBS
149.6 g

FAT
154.5 g

SUGAR
40.8 g

SODIUM
8015.5 mg

BREAKFAST

Vietnamese Crispy Carrot and Cabbage Salad with Sliced Chicken and Chile-Lime Marinade

2 serving



Generated Gemini plan

This plan provides approximately 3,100 calories daily to support muscle gain for a 183cm male. It is strictly gluten-free and dairy-free. To manage blood sugar, it prioritizes low-glycemic complex carbohydrates (quinoa, sweet potatoes, brown rice) and high fiber, while keeping added sugars near zero.

Status:

Hypertrophy focus; Active caloric surplus

Weekly totals

CALORIES
17560

PROTEIN
1188 g

CARBS
1296 g

FAT
862 g

SUGAR
167 g

SODIUM
12365 mg

Day 1

High protein count to support muscle gain for a moderately active male, keeping sugar low to manage blood glucose.

Day 2

Strong focus on protein and omega-3s for inflammation control and muscle growth.

Day 3

Lower sodium day with high volume protein sources.

Day 4

Maintains gluten-free and dairy-free standards while focusing on satiety.

Day 3

Lower sodium day with high volume protein sources.

Day 4

Maintains gluten-free and dairy-free standards while focusing on satiety.

Day 5

Caloric intake balanced with complex carb sources to manage blood sugar.

Day 6

High fat/protein day to support hormone production and muscle repair.

Day 7

Concluding the week with varied protein sources and controlled sugar.

CALORIES
2250

PROTEIN
145 g

CARBS
180 g

FAT
105 g

SUGAR
32 g

SODIUM
1375 mg

BREAKFAST

2

serving

Vietnamese Crispy Carrot and Cabbage Salad with Sliced Chicken and Chile-Lime Marinade

Chicken salad with mint and lime provides a light, high-protein breakfast.

Ingredients

1/4 cup Thai basil
1 1/2 boneless skinless chicken breasts
4 cups cabbage
3 cups carrots
3/4 cup cilantro
1/4 teaspoon crushed red pepper flakes
1/4 cup dry roasted salted peanuts
5 tablespoons fish sauce
7 tablespoons fresh lime juice
1/2 cup fresh mint leaves
2 cloves garlic cloves
peanut oil
1/2 red onion
6 tablespoons rice vinegar
salt
2 scallions
2 serrano chilies
2 shallots
3 tablespoons sugar
1 teaspoon wasabi

432.8 kcal

60.2 g protein

23.6 g carbs

10.2 g fat

FAT

105 g

SUGAR

32 g

SODIUM

1375 mg

BREAKFAST

1 serving

GF Pancakes

Gluten-free pancake alternative for a Sunday breakfast.

Ingredients

1 cup GF oat flour
1 Banana
1 Egg
Almond milk

480 kcal

15 g protein

75 g carbs

12 g fat

LUNCH

1 serving

Roast Beef and Potato Plate

Resistant starch in potatoes helps with blood sugar management.

Ingredients

200g Roast beef
250g Boiled potatoes (chilled/reheated for resistant starch)
Green beans

680 kcal

52 g protein

55 g carbs

28 g fat

LUNCH

1.5 serving

Our Spaghetti Sauce

Beef and sausage sauce bowl for high caloric density.

Ingredients

1 lb Italian sausage
2 tablespoons basil
2 bay leaves
1 carrot
1/2 teaspoon cayenne
2 stalks celery
1 can consomme
5 garlic cloves
1 green pepper
2 teaspoons italian seasoning
3 lbs lean ground beef
3 medium onions
1 teaspoon oregano
2 tablespoons parsley
1/2 teaspoon pepper
1/2 teaspoon sage
1 teaspoon salt
1 can tomato juice
2 cans tomato paste
2 cans tomato sauce
1 can tomato soup
3 cans tomatoes

436.4 kcal

46.8 g protein

9 g carbs

22.7 g fat

SNACK

1 serving

Protein Bar (GF/DF)

Convenience snack to hit macros.

Ingredients

1 Commercial GF/DF Protein bar

250 kcal

20 g protein

25 g carbs

8 g fat

DINNER

1 serving

Pork Tenderloin and Braised Cabbage

Lean pork is excellent for B+ blood types and muscle gain.

Ingredients

200g Pork tenderloin
2 cups Cabbage
1 cup Apple cider vinegar (braising)
1 tbsp Olive oil

550 kcal

48 g protein

15 g carbs

32 g fat

SNACK

1 serving

Almonds

Closing the day with healthy fats.

Ingredients

2 cans tomato sauce
1 can tomato soup
3 cans tomatoes

436.4 kcal46.8 g protein
9 g carbs22.7 g fat

DINNER

2 serving

Anne Rosenzweig\'s Arcadian Eight-Bean Chili

Eight-bean chili provides the fiber and protein needed for recovery.

Ingredients

1 bacon
1/2 ounces beer
1/4 cup cayenne pepper
1/4 cup coriander seeds
1/4 cup cumin seeds
1/4 dried black beans
1/4 dried cranberry beans
1/4 dried navy beans
1/4 dried pink beans
1/4 dried pinto beans
1/4 dried red beans
1/4 dried white beans
2/3 cup garlic
1/4 cup ground cinnamon
5 lean ground beef
1/4 navy beans
5 onions
1/4 cup paprika
1/2 cup poblano peppers

SNACK

1 serving

Almonds

Closing the day with healthy fats.

Ingredients

50g Almonds

290 kcal10 g protein
10 g carbs25 g fat

SNACK

2 serving

Red Beans and Rice

Red beans and rice with sausage for high fat and calories.

Ingredients

1 tablespoon Old Bay Seasoning
1 lb andouille sausages
3 bay leaves
1/2-1 teaspoon cayenne pepper
1 teaspoon celery seeds
cooked rice
2 lbs dried red kidney beans
1/2-1 teaspoon dry crushed red pepper
1 cup dry red wine
1 teaspoon fresh ground black pepper
1 tablespoon fresh oregano
1/2 cup fresh parsley
4 garlic cloves
1 green bell pepper
1 tablespoon gumbo file
1 teaspoon hot sauce
2 onions
1 teaspoon paprika
1 red bell pepper
1 teaspoon salt
1/2 lb salt pork
1 lb smoked beef sausage

3/2 ounces diced tomatoes
1/2 head garlic
2 jalapenos
2 onions
3/2 ounces pinto beans
2 lbs pork
2 teaspoons sea salt
4/6 ounces tomato juice
3/0 ounces tomato sauce
1 lb turkey
2 teaspoons vegetable oil
3 lbs venison
3/0 ounces whole tomatoes

624 kcal

70.9 g protein

21.8 g carbs

29 g fat

Vote on the better plan

Choose the plan that looks more useful, realistic, and safe for the user profile.

RECIPEKG

GEMINI

TIES

TOTAL

Vote on the better plan

Choose the plan that looks more useful, realistic, and safe for the user profile.

RECIPEKG

0

GEMINI

1

TIES

0

TOTAL

1

RecipeKG is better

Gemini is better

Tie

Optional reason for your vote

Submit vote

- Од лева страна може да се види планот изгенериран преку нашиот мултиагентен пристап, а од десна преку директен повик до LLM.
Корисникот има и можност да гласа кој план е подобро изработен за неговите потреби.
- Како агенти и како директен LLM користевме Google Gemini Api.
- Prompt испратен до Gemini за најавениот корисник:

You are a nutrition planning assistant.

Create a practical 7-day nutrition plan directly from the user's profile.
Use normal meal names and ingredient lists that a user could understand.

Return strict JSON only. The JSON must match this exact shape:

```
{
  "goalStatus": "string",
  "summary": "string",
  "days": [
    {
      "day": 1,
      "meals": [
        {
          "slot": "breakfast|lunch|dinner|snack|dessert",
          "recipeName": "string",
          "ingredients": ["string"],
          "servings": 1.0,
          "totalMacros": {
            "calories": 0,
            "protein": 0,
            "carbs": 0,
            "fat": 0,
            "sugar": 0,
            "sodium": 0
          },
          "reason": "string"
        }
      ],
      "totalMacros": {
        "calories": 0,
        "protein": 0,
        "carbs": 0,
        "fat": 0,
        "sugar": 0,
        "sodium": 0
      },
      "rationale": "string"
    }
  ]
}
```

```
}  
],  
"weeklyTotals": {  
  "calories": 0,  
  "protein": 0,  
  "carbs": 0,  
  "fat": 0,  
  "sugar": 0,  
  "sodium": 0  
}  
}
```

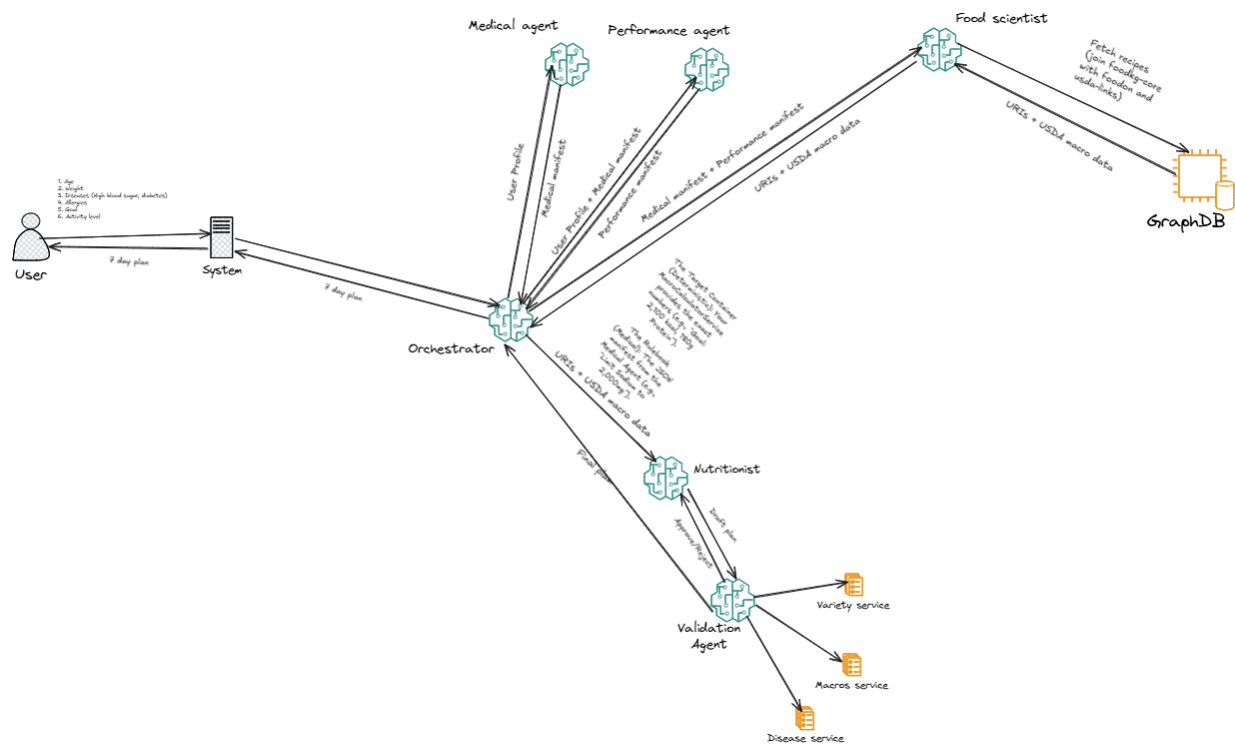
Rules:

1. Return exactly 7 days.
2. Each day should have breakfast, lunch, and dinner. Add snacks only when useful.
3. Respect allergies and diseases/conditions strictly.
4. If the user has diabetes or sugar restriction, keep sugar conservative.
5. If the user has hypertension or sodium restriction, keep sodium conservative.
6. Macros must be realistic estimates for the listed servings.
7. Compute day totalMacros as the sum of meal totalMacros.
8. Compute weeklyTotals as the sum of day totalMacros.
9. Do not include markdown fences or explanation outside JSON.

USER_PROFILE:

Age: 21
Gender: Male
HeightCm: 183.0
WeightKg: 75.0
BloodType: B+
ActivityLevel: Moderate
Goal: Gain muscle
Allergies: [Gluten, Dairy]
DiseasesOrConditions: [High blood sugar]

2. Процесот на генерирање нутриционистички план преку мултиагентен пристап:



- Најпрво корисникот креира профил преку формата:

Create Your Account

Create your profile before accessing any page in the platform.

Register	Login
Name	Surname
Email	Password
Age	Gender
	Select gender ▼
Height (cm)	Weight (kg)
Blood Type	Activity Level
Select blood type ▼	Select activity level ▼
Goal	
Select goal ▼	
Allergies	Diseases
e.g. Peanuts ×	e.g. Diabetes ×
+ Add allergy	+ Add disease
Create Account	

каде внесува сопствени телесни параметри.

- Потоа се испраќа повик до медицинско лице агент кој изгледа:

You are an expert Clinical Dietitian and Medical AI Agent.

Your task is to analyze the user's biological profile and generate a STRICT JSON manifest of dietary constraints.

CRITICAL RULES FOR TERMINOLOGY & STATUS:

1. STATUS DETERMINATION:

- If 'Allergies' is "None" AND 'Diseases/Conditions' is "None", you MUST set "status": "UNCONSTRAINED".
 - ONLY set "status": "CONSTRAINED" if the user has a specific allergy, disease, or clinical condition requiring strict filtering.
2. 'hard_exclusions': Your job is to act as a Keyword Unroller. You must translate the user's dietary restrictions into a flat JSON array of singular, root ingredient keywords and common foods that contain them.
- Do NOT use broad ontology categories. Unroll them into specific, high-risk ingredients.
 - Strip pluralization (e.g., use "peanut" not "peanuts").

EXAMPLES:

- User: "I have a fish allergy." -> Output: ["fish", "salmon", "tuna", "cod", "trout"]
- User: "I have a dairy allergy" -> Output: ["milk", "cheese", "butter", "whey"]
- Uer: "I have a gluten allergy" -> Output: ["wheat", "bread", "flour", "pasta", "crouton", "pretzel"]

3. 'nutrient_caps' RULES:

- Each entry must contain exactly one USDA nutrient name.
- Do NOT combine nutrients (no commas, no "and", no grouping).
- Do NOT repeat the same nutrient more than once.
- Use canonical USDA names only (examples: "Carbs", "Sugar", "Sodium", "Energy", "Protein", "Fat").

4. IF UNCONSTRAINED: Leave the 'hard_exclusions', 'nutrient_caps', and 'required_boosts' arrays completely EMPTY.

USER PROFILE:

Age: 21

Gender: Male

Height (cm): 183.0

Weight (kg): 75.0

Blood Type: B+

Activity Level: Moderate

Goal: Gain muscle

Allergies: [Gluten, Dairy]

Diseases/Conditions: [High blood sugar]

EXPECTED JSON SCHEMA:

```

{
  "status": "CONSTRAINED",
  "constraints": {
    "hard_exclusions": ["<String>"],
    "nutrient_caps": [
      { "nutrient": "<String>", "max_value": <Number>, "unit": "<String>" }
    ]
  },
  "medical_rationale": "<String detailing the clinical reasoning>"
}

```

- Додека одговорот е:

```

{
  "status": "CONSTRAINED",
  "constraints": {
    "hard_exclusions": [
      "wheat",
      "bread",
      "flour",
      "pasta",
      "crouton",
      "pretzel",
      "barley",
      "rye",
      "semolina",
      "couscous",
      "bulgur",
      "milk",
      "cheese",
      "butter",
      "cream",
      "yogurt",
      "whey",
      "casein"
    ],
    "nutrient_caps": [
      {
        "nutrient": "Sugar",

```

```

      "max_value": 45,
      "unit": "g"
    },
    {
      "nutrient": "Carbs",
      "max_value": 350,
      "unit": "g"
    }
  ]
},
"medical_rationale": "The user has gluten and dairy allergies, so common
gluten-containing grains and dairy-derived ingredients are excluded. High blood
sugar requires limiting sugar and moderating total carbohydrate intake while still
supporting the user's muscle-gain goal."
}

```

- Потоа се испраќа повик до „Performance“ агентот, во кој се содржат податоците за корисникот, како и генерираниот „медицински манифест“. Неговата цел е да одреди граници за внес на нутритивни вредности врз база на целта на корисникот, како и препораките од медицинскиот агент.
- Пример на prompt:

You are a sports nutrition performance agent.

Calculate performance nutrition targets for this user and return STRICT JSON only. These targets are performance-based, not medical hard constraints. Use the user's goal, body metrics, and activity level to estimate daily calories, protein, carbs, and fat. Then derive per-serving targets suitable for ranking individual recipes. Respect the medical manifest when choosing reasonable upper/lower ranges, but do not duplicate medical hard exclusions.

Return exactly this shape:

```

{
  "goalStatus": "HYPERTROPHY|FAT_LOSS|MAINTENANCE|ENDURANCE",

```

```

"dailyTargets": [
  {"nutrient": "calories", "min": 0, "target": 0, "max": 0, "unit": "kcal", "period": "day"},
  {"nutrient": "protein", "min": 0, "target": 0, "max": 0, "unit": "g", "period": "day"},
  {"nutrient": "carbs", "min": 0, "target": 0, "max": 0, "unit": "g", "period": "day"},
  {"nutrient": "fat", "min": 0, "target": 0, "max": 0, "unit": "g", "period": "day"}
],
"perServingTargets": [
  {"nutrient": "calories", "min": 0, "target": 0, "max": 0, "unit": "kcal", "period": "serving"},
  {"nutrient": "protein", "min": 0, "target": 0, "max": 0, "unit": "g", "period": "serving"},
  {"nutrient": "carbs", "min": 0, "target": 0, "max": 0, "unit": "g", "period": "serving"},
  {"nutrient": "fat", "min": 0, "target": 0, "max": 0, "unit": "g", "period": "serving"}
],
"rationale": "short explanation"
}

```

PROFILE:

Age: 21

Gender: Male

WeightKg: 75.00

HeightCm: 183.00

Goal: Gain muscle

ActivityLevel: Moderate

MEDICAL MANIFEST:

```

{"status": "CONSTRAINED", "constraints": {"hard_exclusions": ["wheat", "bread", "flour", "pa
sta", "crouton", "pretzel", "barley", "rye", "semolina", "couscous", "milk", "cheese", "butter", "cre
am", "yogurt", "whey", "casein"], "nutrient_caps": [{"nutrient": "Sugar", "max_value": 45, "unit":
"g"}, {"nutrient": "Carbs", "max_value": 350, "unit": "g"}]}, "medical_rationale": "The user has
gluten and dairy allergies, so common gluten-containing grains and dairy-derived
ingredients are excluded. High blood sugar requires limiting sugar and moderating total
carbohydrate intake while still supporting the user's muscle-gain goal."}

```

- Пример на одговор:

```

{
  "goalStatus": "HYPERTROPHY",
  "dailyTargets": [
    {
      "nutrient": "calories",

```

```
    "min": 2850,
    "target": 3100,
    "max": 3350,
    "unit": "kcal",
    "period": "day"
  },
  {
    "nutrient": "protein",
    "min": 135,
    "target": 165,
    "max": 210,
    "unit": "g",
    "period": "day"
  },
  {
    "nutrient": "carbs",
    "min": 240,
    "target": 320,
    "max": 350,
    "unit": "g",
    "period": "day"
  },
  {
    "nutrient": "fat",
    "min": 80,
    "target": 105,
    "max": 135,
    "unit": "g",
    "period": "day"
  }
],
"perServingTargets": [
  {
    "nutrient": "calories",
    "min": 500,
    "target": 750,
    "max": 950,
    "unit": "kcal",
    "period": "serving"
  },
  {
```

```

{
  "nutrient": "protein",
  "min": 25,
  "target": 40,
  "max": 60,
  "unit": "g",
  "period": "serving"
},
{
  "nutrient": "carbs",
  "min": 45,
  "target": 75,
  "max": 100,
  "unit": "g",
  "period": "serving"
},
{
  "nutrient": "fat",
  "min": 15,
  "target": 25,
  "max": 40,
  "unit": "g",
  "period": "serving"
}
],
"rationale": "The user is a moderately active 21-year-old male aiming to gain muscle,
so the plan targets a caloric surplus with high protein intake to support hypertrophy.
Carbohydrates are kept below the medical cap due to high blood sugar, while fats are
increased moderately to help meet calorie needs without excessive carbohydrate load."
}

```

- Следниот чекор е избор на соодветни рецепти од нашата граф база на податоци. За тоа е задолжен сервис „научник за храна“.
 - Најпрво се конструира прашалник за избор на рецепти преку кој се филтрираат рецептите кои содржат состојки забранети од медицинскиот агент.
- Пример на прашалник:

```

PREFIX heals: <http://idea.rpi.edu/heals/kb/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX usda: <http://example.org/usda/>
PREFIX foodon: <http://purl.obolibrary.org/obo/>
PREFIX obolInOwl: <http://www.geneontology.org/formats/obolInOwl#>

```

```

SELECT DISTINCT ?recipe ?recipeLabel ?use ?ingName ?ingLabel ?qty
?unit (SAMPLE(?usdaUri) AS ?usdaUri)
WHERE {
  {
    SELECT ?recipe ?recipeLabel (COUNT(DISTINCT ?usdaItem) AS
?mappedCount) WHERE {
      ?recipe a heals:recipe ;
      rdfs:label ?recipeLabel .

```

```

FILTER(!
  CONTAINS(LCASE(STR(?recipeLabel)), "wheat") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "bread") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "flour") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "pasta") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "crouton") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "pretzel") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "barley") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "rye") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "semolina") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "couscous") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "milk") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "cheese") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "butter") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "cream") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "yogurt") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "whey") ||
  CONTAINS(LCASE(STR(?recipeLabel)), "casein")
))

```

```

FILTER NOT EXISTS {
  ?recipe heals:uses/heals:ing_name ?badName .

```



```

FILTER(
  CONTAINS(LCASE(STR(?badName)), "wheat") ||
  CONTAINS(LCASE(STR(?badName)), "bread") ||
  CONTAINS(LCASE(STR(?badName)), "flour") ||
  CONTAINS(LCASE(STR(?badName)), "pasta") ||
  CONTAINS(LCASE(STR(?badName)), "crouton") ||
  CONTAINS(LCASE(STR(?badName)), "pretzel") ||
  CONTAINS(LCASE(STR(?badName)), "barley") ||
  CONTAINS(LCASE(STR(?badName)), "rye") ||
  CONTAINS(LCASE(STR(?badName)), "semolina") ||
  CONTAINS(LCASE(STR(?badName)), "couscous") ||
  CONTAINS(LCASE(STR(?badName)), "milk") ||
  CONTAINS(LCASE(STR(?badName)), "cheese") ||
  CONTAINS(LCASE(STR(?badName)), "butter") ||
  CONTAINS(LCASE(STR(?badName)), "cream") ||
  CONTAINS(LCASE(STR(?badName)), "yogurt") ||
  CONTAINS(LCASE(STR(?badName)), "whey") ||
  CONTAINS(LCASE(STR(?badName)), "casein")
)
}

```

```

FILTER NOT EXISTS {
  ?recipe heals:uses/heals:ing_name ?name .
  ?name owl:equivalentClass ?foodOnClass .
  FILTER(STRSTARTS(STR(?foodOnClass),
"http://purl.obolibrary.org/obo/FOODON_"))

  ?foodOnClass rdfs:subClassOf* ?parentClass .
  FILTER(STRSTARTS(STR(?parentClass),
"http://purl.obolibrary.org/obo/FOODON_"))

  OPTIONAL {
    ?parentClass rdfs:label ?foLabel .
    FILTER(LANG(?foLabel) = "" || LANGMATCHES(LANG(?foLabel),
"en"))
  }

  OPTIONAL {
    ?parentClass obolnOwl:hasSynonym ?foSyn .

```

```

    }

    FILTER(
        REGEX(LCASE(STR(?foLabel)),
            "(^|\\W)(wheat|bread|flour|pasta|crouton|pretzel|barley|rye|semolina|couscous|mil
            k|cheese|butter|cream|yogurt|whey|casein)(\\W|$)" ||
        REGEX(LCASE(STR(?foSyn)),
            "(^|\\W)(wheat|bread|flour|pasta|crouton|pretzel|barley|rye|semolina|couscous|mil
            k|cheese|butter|cream|yogurt|whey|casein)(\\W|$)"
        )
    )
}

```

```

OPTIONAL {
    ?recipe heals:uses/heals:ing_name ?candidateIng .
    ?candidateIng owl:equivalentClass ?usdaltem .
    FILTER(CONTAINS(STR(?usdaltem), "fdc.nal.usda.gov"))
}
}

```

```

GROUP BY ?recipe ?recipeLabel
ORDER BY DESC(?mappedCount) (LCASE(STR(?recipeLabel)))
LIMIT 500
}

```

```

?recipe heals:uses ?use .
?use heals:ing_name ?ingName .

```

```

OPTIONAL {
    ?ingName rdfs:label ?ingLabel
}

```

```

OPTIONAL {
    ?use heals:ing_quantity ?qty
}

```

```

OPTIONAL {
    ?use heals:ing_unit ?unit
}

```

```

OPTIONAL {

```

```

    ?ingName owl:equivalentClass ?usdaltem .
    FILTER(CONTAINS(STR(?usdaltem), "fdc.nal.usda.gov"))
    BIND(STR(?usdaltem) AS ?usdaUrl)
  }
}

```

GROUP BY ?recipe ?recipeLabel ?use ?ingName ?ingLabel ?qty ?unit

- Потоа се пополнуваат нутритивните вредности на секој од рецептите преку USDA Api и се филтрираат рецептите кои ги надминуваат граничните нутритивни вредности поставени од медицинскиот агент.
- Резултатот кој се враќа изгледа:

```

{
  "recipes": [
    {
      "uri":
        "http://idea.rpi.edu/heals/kb/recipe/f3da3d40-Vietnamese%20Crispy%20C
        arrot%20and%20Cabbage%20Salad%20with%20Sliced%20Chicken%20
        and%20Chile-Lime%20Marinade",
      "label": "Vietnamese Crispy Carrot and Cabbage Salad with Sliced
        Chicken and Chile-Lime Marinade",
      "servings": 4.0,
      "usdaIngredientIds": [
        "https://fdc.nal.usda.gov/fdc-app.html#/food-details/171077",
        "https://fdc.nal.usda.gov/fdc-app.html#/food-details/170393",
        "https://fdc.nal.usda.gov/fdc-app.html#/food-details/169988"
      ],
      "usdaIngredientText": "skinless chicken breasts, carrots, cabbage,
        basil, cilantro, lime juice, fish sauce, ginger, garlic",
      "calories": 211.4,
      "protein": 30.1,
      "carbs": 11.8,
      "fat": 5.1,
      "sugar": 5.4,
      "sodium": 590.0,
      "performanceScore": 87.5,
    }
  ]
}

```

```
"performanceReason": "High protein and moderate calories make this
useful for hypertrophy while keeping carbohydrates controlled.",
"macroUnits": {
  "calories": "kcal per serving",
  "protein": "g per serving",
  "carbs": "g per serving",
  "fat": "g per serving",
  "sugar": "g per serving",
  "sodium": "mg per serving"
},
"ingredients": [
  {
    "name": "thai basil",
    "quantity": "1/4",
    "unit": "cup",
    "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/172232",
    "macroResolved": true,
    "macroResolutionReason": "Matched to USDA basil."
  },
  {
    "name": "boneless skinless chicken breasts",
    "quantity": "1 1/2",
    "unit": "lb",
    "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/171077",
    "macroResolved": true,
    "macroResolutionReason": "Matched to USDA chicken breast."
  },
  {
    "name": "cabbage",
    "quantity": "4",
    "unit": "cups",
    "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/169975",
    "macroResolved": true,
    "macroResolutionReason": "Matched to USDA cabbage."
  },
  {
    "name": "carrots",
```

```
    "quantity": "3",
    "unit": "cups",
    "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/170393",
    "macroResolved": true,
    "macroResolutionReason": "Matched to USDA carrots."
  },
  {
    "name": "cilantro",
    "quantity": "3/4",
    "unit": "cup",
    "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/169997",
    "macroResolved": true,
    "macroResolutionReason": "Matched to USDA cilantro."
  },
  {
    "name": "crushed red pepper flakes",
    "quantity": "1/4",
    "unit": "teaspoon",
    "usdaUrl": null,
    "macroResolved": false,
    "macroResolutionReason": "Spice amount is nutritionally negligible."
  },
  {
    "name": "roasted salted peanuts",
    "quantity": "1/4",
    "unit": "cup",
    "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/172430",
    "macroResolved": true,
    "macroResolutionReason": "Matched to USDA peanuts."
  },
  {
    "name": "fish sauce",
    "quantity": "5",
    "unit": "tablespoons",
    "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/174276",
    "macroResolved": true,
```

```

        "macroResolutionReason": "Matched to USDA fish sauce."
      },
      {
        "name": "fresh lime juice",
        "quantity": "7",
        "unit": "tablespoons",
        "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/169917",
        "macroResolved": true,
        "macroResolutionReason": "Matched to USDA lime juice."
      },
      {
        "name": "mint leaves",
        "quantity": "1/2",
        "unit": "cup",
        "usdaUrl":
"https://fdc.nal.usda.gov/fdc-app.html#/food-details/173475",
        "macroResolved": true,
        "macroResolutionReason": "Matched to USDA mint."
      }
    ]
  },
  {
    "uri":
"http://idea.rpi.edu/heals/kb/recipe/85654f42-Our%20Spaghetti%20Sauce"
    ,
    "label": "Our Spaghetti Sauce",
    "servings": 8.0,
    "usdaIngredientIds": [
      "https://fdc.nal.usda.gov/fdc-app.html#/food-details/173867",
      "https://fdc.nal.usda.gov/fdc-app.html#/food-details/170457",
      "https://fdc.nal.usda.gov/fdc-app.html#/food-details/170000"
    ],
    "usdaIngredientText": "italian sausage, tomato sauce, tomato paste,
tomatoes, onion, garlic, oregano, parsley",
    "calories": 290.9,
    "protein": 31.2,
    "carbs": 6.0,
    "fat": 15.1,
    "sugar": 4.8,

```

```

"sodium": 720.0,
"performanceScore": 82.0,
"performanceReason": "High protein and calorie density make this
useful for muscle gain, though sodium should be watched.",
"macroUnits": {
  "calories": "kcal per serving",
  "protein": "g per serving",
  "carbs": "g per serving",
  "fat": "g per serving",
  "sugar": "g per serving",
  "sodium": "mg per serving"
},
"
.....

```

- Потоа податоците за корисникот, „медицинскиот манифест“, „performance manifest“ и рецептите се испраќаат до агентот нутриционист. Процесот се одвива во три чекори, каде во секој чекор се испраќаат по 30 рецепти (сумаризирани), како и меморијата од претходниот чекор.
- Пример за барање и одговор во вториот чекор:

Барање (prompt):

You are a nutritionist recipe scout. Maintain a rolling shortlist; do not build the 7-day plan yet.

Rules: use only IDs in CURRENT_SHORTLIST or NEW_BATCH; max 24 selected IDs; respect medical caps/exclusions; prefer goal fit, variety, usable macros, and meal-slot coverage. Keep 5-7 breakfast-suitable recipes when available. Judge breakfast suitability from label, ingredients, preparation style, and practical context. Do not choose breakfast coverage only by macro density. USDA macros are estimates; note implausible values or important unresolved ingredients.

USER_PROFILE:

```

{"age":21,"gender":"Male","heightCm":183.0,"weightKg":75.0,"activityLevel":"Moderate",
"goal":"Gain muscle","allergies":["Gluten","Dairy"],"diseases":["High blood sugar"]}

```

MEDICAL:

```
{"status":"CONSTRAINED","exclude":["wheat","bread","flour","pasta","crouton","pretzel","barley","rye","semolina","couscous","milk","cheese","butter","cream","yogurt","whey","casein"],"caps":[{"nutrient":"Sugar","max":45.0,"unit":"g"},{"nutrient":"Carbs","max":350.0,"unit":"g"}],"boosts":[]}
```

TARGETS:

```
{"goal":"HYPERTROPHY","daily":{"calories":{"min":2850.0,"target":3100.0,"max":3350.0,"unit":"kcal"},"protein":{"min":135.0,"target":165.0,"max":210.0,"unit":"g"},"carbs":{"min":240.0,"target":320.0,"max":350.0,"unit":"g"},"fat":{"min":80.0,"target":105.0,"max":135.0,"unit":"g"}}}
```

BATCH_NUMBER:

2

CURRENT_SELECTION_MEMORY:

```
{"selectedRecipeIds":["R1","R2","R4","R7","R8","R9","R10","R11","R12","R13","R15","R16","R18","R19","R20","R21","R22","R24","R25","R26","R27","R28","R31","R35"],"missingNeeds":["Still lacks breakfast-suitable recipes. Requires gluten-free, dairy-free breakfast options like egg-based dishes, smoothies, or oatmeal alternatives","Limited variety in seafood-based recipes due to unresolved allergen concerns"],"nutritionConcerns":["R31 has high sodium (1137mg/serving), requiring portion control to avoid exceeding daily limits","R32 contains implausibly high sodium (10188mg/serving) – likely data error","R46 and R36 exceed 18g sugar/serving, nearing daily cap","Multiple recipes (R40, R57, R48) have sodium >1800mg/serving, posing risk for blood sugar management"],"planningNotes":["R35 (Ropa Vieja) offers lean beef and rice for balanced macros, suitable as a lighter main dish","R31 (Beef Wraps) provides high protein with lettuce-based carbs, adaptable for meal prep","No breakfast candidates identified in NEW_BATCH; prioritize egg-based or smoothie recipes in next batch"]}
```

CURRENT_SHORTLIST:

```
[  
  {  
    "id": "R1",  
    "label": "Taqueria Style Tacos-Carne Asada",  
    "macros": {  
      "calories": 720.0,  
      "protein": 52.0,  
      "carbs": 62.0,
```



```
    "fat": 28.0,  
    "sugar": 7.0,  
    "sodium": 980.0  
  },  
  "servings": 4.0,  
  "ingredients": [  
    "beef",  
    "corn tortilla",  
    "lime",  
    "cilantro"  
  ]  
},  
{  
  "id": "R2",  
  "label": "Ropa Vieja",  
  "macros": {  
    "calories": 690.0,  
    "protein": 48.0,  
    "carbs": 76.0,  
    "fat": 22.0,  
    "sugar": 9.0,  
    "sodium": 940.0  
  },  
  "servings": 4.0,  
  "ingredients": [  
    "beef",  
    "rice",  
    "tomato",  
    "pepper"  
  ]  
},  
{  
  "id": "R4",  
  "label": "Chicken and Rice Skillet",  
  "macros": {  
    "calories": 710.0,  
    "protein": 55.0,  
    "carbs": 74.0,  
    "fat": 21.0,  
    "sugar": 6.0,
```

```
    "sodium": 830.0
  },
  "servings": 4.0,
  "ingredients": [
    "chicken",
    "rice",
    "pepper",
    "olive oil"
  ]
},
{
  "id": "R7",
  "label": "Turkey Bean Chili",
  "macros": {
    "calories": 650.0,
    "protein": 50.0,
    "carbs": 66.0,
    "fat": 19.0,
    "sugar": 8.0,
    "sodium": 890.0
  },
  "servings": 6.0,
  "ingredients": [
    "turkey",
    "bean",
    "tomato",
    "onion"
  ]
},
{
  "id": "R31",
  "label": "Asian Beef & Lettuce Wraps",
  "macros": {
    "calories": 610.0,
    "protein": 46.0,
    "carbs": 48.0,
    "fat": 24.0,
    "sugar": 6.0,
    "sodium": 1137.0
  },
}
```

```
"servings": 4.0,
"ingredients": [
  "beef",
  "lettuce",
  "rice",
  "ginger"
]
},
{
  "id": "R35",
  "label": "Ropa Vieja",
  "macros": {
    "calories": 650.0,
    "protein": 49.0,
    "carbs": 70.0,
    "fat": 20.0,
    "sugar": 3.3,
    "sodium": 890.0
  },
  "servings": 4.0,
  "ingredients": [
    "beef",
    "rice",
    "tomato",
    "pepper"
  ]
}
]
```

NEW_BATCH:

```
[
  {
    "id": "R41",
    "label": "Berry Chia Breakfast Bowl",
    "macros": {
      "calories": 520.0,
      "protein": 24.0,
      "carbs": 72.0,
      "fat": 18.0,
      "sugar": 13.0,
```

```
    "sodium": 160.0
  },
  "servings": 2.0,
  "ingredients": [
    "chia",
    "berries",
    "banana",
    "almond milk",
    "oats"
  ]
},
{
  "id": "R43",
  "label": "Sweet Potato Egg Hash",
  "macros": {
    "calories": 610.0,
    "protein": 32.0,
    "carbs": 68.0,
    "fat": 24.0,
    "sugar": 10.0,
    "sodium": 520.0
  },
  "servings": 4.0,
  "ingredients": [
    "egg",
    "sweet potato",
    "spinach",
    "olive oil",
    "pepper"
  ]
},
{
  "id": "R44",
  "label": "Chicken Rice Bowl",
  "macros": {
    "calories": 680.0,
    "protein": 54.0,
    "carbs": 72.0,
    "fat": 18.0,
    "sugar": 7.0,
```

```
    "sodium": 710.0
  },
  "servings": 4.0,
  "ingredients": [
    "chicken",
    "rice",
    "pepper",
    "onion",
    "olive oil"
  ]
},
{
  "id": "R46",
  "label": "Mango Chicken Rice",
  "macros": {
    "calories": 640.0,
    "protein": 43.0,
    "carbs": 82.0,
    "fat": 17.0,
    "sugar": 19.5,
    "sodium": 760.0
  },
  "servings": 4.0,
  "ingredients": [
    "chicken",
    "rice",
    "mango",
    "pepper",
    "lime"
  ]
},
{
  "id": "R48",
  "label": "Smoked Sausage Rice Bowl",
  "macros": {
    "calories": 760.0,
    "protein": 38.0,
    "carbs": 78.0,
    "fat": 34.0,
    "sugar": 8.0,
```

```
    "sodium": 1840.0
  },
  "servings": 4.0,
  "ingredients": [
    "sausage",
    "rice",
    "pepper",
    "onion",
    "tomato"
  ]
}
```

OUTPUT_SCHEMA:

```
{
  "selectedRecipeIds": ["R1"],
  "rejectedRecipeIds": ["R2"],
  "substitutions": [
    {
      "removedRecipeId": "R3",
      "addedRecipeId": "R31",
      "reason": "short reason"
    }
  ],
  "missingNeeds": ["short notes about what the shortlist still lacks, including breakfast gaps"],
  "nutritionConcerns": ["short notes about sodium, implausible macros, unresolved important ingredients, etc."],
  "planningNotes": ["short notes useful for the final plan builder, including which selected recipes seem most breakfast-suitable"]
}
```

Одговор:

```
{
  "selectedRecipeIds": [
    "R1",
    "R2",
```

```
"R4",
"R7",
"R8",
"R9",
"R10",
"R11",
"R12",
"R13",
"R15",
"R16",
"R18",
"R19",
"R20",
"R21",
"R22",
"R24",
"R25",
"R31",
"R35",
"R41",
"R43",
"R44"
],
"rejectedRecipelds": [
  "R26",
  "R27",
  "R28",
  "R46",
  "R48"
],
"substitutions": [
  {
    "removedRecipeld": "R27",
    "addedRecipeld": "R41",
    "reason": "Added Berry Chia Breakfast Bowl because it is clearly breakfast-suitable, low sodium, and useful for carbohydrate intake without exceeding the sugar cap alone."
  },
  {
    "removedRecipeld": "R28",
    "addedRecipeld": "R43",
```

"reason": "Added Sweet Potato Egg Hash as a gluten-free, dairy-free breakfast option with stronger protein and moderate sodium."

},

{

"removedRecipeId": "R26",

"addedRecipeId": "R44",

"reason": "Added Chicken Rice Bowl because it provides high protein and carbohydrate density with lower sodium than several existing savory options."

}

],

"missingNeeds": [

"The shortlist now has stronger breakfast coverage from R41 and R43, but may still need one more low-sodium breakfast or snack option to avoid repeating them too often across 7 days.",

"High-calorie options are available, but sodium must be managed carefully when using R31, R15, or other savory meat-based recipes.",

"Sugar-sensitive days should avoid pairing R41 with fruit-heavy or dessert-like recipes."

],

"nutritionConcerns": [

"R31 remains useful for protein but has high sodium at 1137mg per serving, so it should usually be limited to 0.5-1.0 serving.",

"R46 has 19.5g sugar per serving, which is not unsafe alone but makes the 45g daily sugar cap harder to satisfy.",

"R48 has 1840mg sodium per serving, leaving very little sodium room for the rest of the day.",

"R41 is breakfast-appropriate but contributes 13g sugar, so it should be paired with low-sugar meals.",

"R43 is one of the best breakfast candidates because it includes eggs and sweet potato while staying moderate in sodium and sugar."

],

"planningNotes": [

"Preferred breakfast anchors: R41 and R43.",

"Use R44 as a flexible lunch or dinner anchor for hypertrophy because it has high protein, good carbohydrates, and manageable sodium.",

"Use R1, R2, R4, R7, R35, and R44 as main-meal anchors.",

"Limit R31 to smaller servings or pair it with very low-sodium breakfast/snack choices.",

"Avoid R48 unless the final plan has an otherwise very low-sodium day."

]

}

- Потоа се испраќа барање до агентот да го направи 7 - дневниот план:

You are a certified sports nutritionist. Build a practical 7-day plan using ONLY candidate IDs. JSON only. Do not invent recipes/ingredients. USDA macros are estimates; use nutrition judgment for implausible values.

USER_PROFILE:

```
{"age":21,"gender":"Male","heightCm":183.0,"weightKg":75.0,"activityLevel":"Moderate","goal":"Gain muscle","allergies":["Gluten","Dairy"],"diseases":["High blood sugar"]}
```

MEDICAL:

```
{"status":"CONSTRAINED","exclude":["wheat","bread","flour","pasta","crouton","pretzel","barley","rye","semolina","couscous","milk","cheese","butter","cream","yogurt","whey","casein"],"caps":[{"nutrient":"Sugar","max":45.0,"unit":"g"},{"nutrient":"Carbs","max":350.0,"unit":"g"}],"boosts":[]}
```

TARGETS:

```
{"goal":"HYPERTROPHY","daily":{"calories":{"min":2850.0,"target":3100.0,"max":3350.0,"unit":"kcal"},"protein":{"min":135.0,"target":165.0,"max":210.0,"unit":"g"},"carbs":{"min":240.0,"target":320.0,"max":350.0,"unit":"g"},"fat":{"min":80.0,"target":105.0,"max":135.0,"unit":"g"}}}
```

AVAILABLE_RECIPE_CANDIDATES:

```
[  
  {  
    "id": "R1",  
    "label": "Taqueria Style Tacos-Carne Asada",  
    "macros": {  
      "calories": 720.0,  
      "protein": 52.0,  
      "carbs": 62.0,  
      "fat": 28.0,  
      "sugar": 7.0,  
      "sodium": 980.0  
    },  
  },  
]
```

```
"servings": 4.0,
"ingredients": [
  "beef",
  "corn tortilla",
  "lime",
  "cilantro",
  "onion"
]
},
{
  "id": "R2",
  "label": "Ropa Vieja",
  "macros": {
    "calories": 690.0,
    "protein": 48.0,
    "carbs": 76.0,
    "fat": 22.0,
    "sugar": 9.0,
    "sodium": 940.0
  },
  "servings": 4.0,
  "ingredients": [
    "beef",
    "rice",
    "tomato",
    "pepper",
    "onion"
  ]
},
{
  "id": "R4",
  "label": "Chicken and Rice Skillet",
  "macros": {
    "calories": 710.0,
    "protein": 55.0,
    "carbs": 74.0,
    "fat": 21.0,
    "sugar": 6.0,
    "sodium": 830.0
  },
  },
```

```
"servings": 4.0,
"ingredients": [
  "chicken",
  "rice",
  "pepper",
  "olive oil",
  "tomato"
]
},
{
  "id": "R7",
  "label": "Turkey Bean Chili",
  "macros": {
    "calories": 650.0,
    "protein": 50.0,
    "carbs": 66.0,
    "fat": 19.0,
    "sugar": 8.0,
    "sodium": 890.0
  },
  "servings": 6.0,
  "ingredients": [
    "turkey",
    "bean",
    "tomato",
    "onion",
    "chili"
  ]
},
{
  "id": "R31",
  "label": "Asian Beef & Lettuce Wraps",
  "macros": {
    "calories": 610.0,
    "protein": 46.0,
    "carbs": 48.0,
    "fat": 24.0,
    "sugar": 6.0,
    "sodium": 1137.0
  },
  },
```

```
"servings": 4.0,
"ingredients": [
  "beef",
  "lettuce",
  "rice",
  "ginger",
  "tamari"
]
},
{
  "id": "R35",
  "label": "Ropa Vieja",
  "macros": {
    "calories": 650.0,
    "protein": 49.0,
    "carbs": 70.0,
    "fat": 20.0,
    "sugar": 3.3,
    "sodium": 890.0
  },
  "servings": 4.0,
  "ingredients": [
    "beef",
    "rice",
    "tomato",
    "pepper",
    "onion"
  ]
},
{
  "id": "R41",
  "label": "Berry Chia Breakfast Bowl",
  "macros": {
    "calories": 520.0,
    "protein": 24.0,
    "carbs": 72.0,
    "fat": 18.0,
    "sugar": 13.0,
    "sodium": 160.0
  },
}
```

```
"servings": 2.0,  
"ingredients": [  
  "chia",  
  "berries",  
  "banana",  
  "almond milk",  
  "oats"  
]  
},  
{  
  "id": "R43",  
  "label": "Sweet Potato Egg Hash",  
  "macros": {  
    "calories": 610.0,  
    "protein": 32.0,  
    "carbs": 68.0,  
    "fat": 24.0,  
    "sugar": 10.0,  
    "sodium": 520.0  
  },  
  "servings": 4.0,  
  "ingredients": [  
    "egg",  
    "sweet potato",  
    "spinach",  
    "olive oil",  
    "pepper"  
  ]  
},  
{  
  "id": "R44",  
  "label": "Chicken Rice Bowl",  
  "macros": {  
    "calories": 680.0,  
    "protein": 54.0,  
    "carbs": 72.0,  
    "fat": 18.0,  
    "sugar": 7.0,  
    "sodium": 710.0  
  },  
}
```

```

"servings": 4.0,
"ingredients": [
  "chicken",
  "rice",
  "pepper",
  "onion",
  "olive oil"
]
}
]

```

RULES:

1. Exactly 7 days; 3-5 meals/day; include breakfast/lunch/dinner daily.
2. Breakfast: choose recipes you judge breakfast-suitable from label, ingredients, preparation style, and context when available. Do not choose breakfast solely for macro density.
3. Medical constraints are mandatory.
4. Servings allowed: 0.5, 1.0, 1.5, 2.0.
5. Max 2 uses per recipe ID/week; count before answering.
6. Add snack/dessert only when useful.
7. Output recipe IDs only, not full recipe objects.

OUTPUT_SCHEMA:

```

{
  "goalStatus": "string",
  "summary": "string",
  "days": [
    {
      "day": 1,
      "meals": [
        {
          "slot": "breakfast|lunch|dinner|snack|dessert",
          "recipeId": "R1",
          "servings": 1.0,
          "reason": "short practical reason"
        }
      ],
      "dayRationale": "short rationale"
    }
  ]
}
]

```

```
}
```

TASK:

Build the final 7-day plan. Use at least 18 unique recipe IDs when 18+ are available, otherwise use as many distinct IDs as practical. Use fewer only for medical constraints. Count recipe usage before answering; max 2 uses per recipe/week.

- И одговор (скратен на еден ден само):

```
{
  "goalStatus": "HYPERTROPHY",
  "summary": "One-day example from the 7-day hypertrophy plan using only
selected recipe IDs. The day prioritizes high protein and calories while keeping
gluten and dairy exclusions in mind.",
  "days": [
    {
      "day": 7,
      "meals": [
        {
          "slot": "breakfast",
          "recipeId": "R66",
          "servings": 2.0,
          "reason": "Vietnamese Crispy Carrot and Cabbage Salad with Sliced
Chicken and Chile-Lime Marinade provides a light but protein-rich breakfast
option."
        },
        {
          "slot": "lunch",
          "recipeId": "R56",
          "servings": 1.5,
          "reason": "Our Spaghetti Sauce adds a calorie-dense beef and sausage
meal for midday energy."
        },
        {
          "slot": "dinner",
          "recipeId": "R72",
          "servings": 2.0,
```

```

      "reason": "Anne Rosenzweig's Arcadian Eight-Bean Chili provides fiber
and protein for recovery."
    },
    {
      "slot": "snack",
      "recipeId": "R48",
      "servings": 2.0,
      "reason": "Red Beans and Rice adds additional calories, carbohydrates,
and protein to support hypertrophy targets."
    }
  ],
  "dayRationale": "This day uses the four screenshot recipes as a high-calorie,
high-protein sample day. Breakfast is lighter and chicken-based, lunch and
dinner provide dense savory meals, and the snack helps push calories toward
the muscle-gain target."
}
]
}

```

- Потоа планот поминува низ неколку итерации на валидација на разновидност на рецепти (не смее еден рецепт да се појавува повеќе од два пати неделно) и дали се задоволени границите поставени од медицинскиот и „performance“ агентите.
- Пример за prompt до LLM за поправка на планот:

You are a certified sports nutritionist. Build a practical 7-day plan using ONLY candidate IDs. JSON only. Do not invent recipes/ingredients. USDA macros are estimates; use nutrition judgment for implausible values.

USER_PROFILE:

```

{"age":21,"gender":"Male","heightCm":183.0,"weightKg":75.0,"activityLevel":"Moderate",
"goal":"Gain muscle","allergies":["Gluten","Dairy"],"diseases":["High blood sugar"]}

```

MEDICAL:


```
{"status":"CONSTRAINED","exclude":["wheat","bread","flour","pasta","crouton","pretzel","barley","rye","semolina","couscous","milk","cheese","butter","cream","yogurt","whey","casein"],"caps":[{"nutrient":"Sugar","max":45.0,"unit":"g"},{"nutrient":"Carbs","max":350.0,"unit":"g"}],"boosts":[]}
```

TARGETS:

```
{"goal":"HYPERTROPHY","daily":{"calories":{"min":2850.0,"target":3100.0,"max":3350.0,"unit":"kcal"},"protein":{"min":135.0,"target":165.0,"max":210.0,"unit":"g"},"carbs":{"min":240.0,"target":320.0,"max":350.0,"unit":"g"},"fat":{"min":80.0,"target":105.0,"max":135.0,"unit":"g"}}}
```

AVAILABLE_RECIPE_CANDIDATES:

```
[  
  {  
    "id": "R1",  
    "label": "Taqueria Style Tacos-Carne Asada",  
    "macros": {  
      "calories": 720.0,  
      "protein": 52.0,  
      "carbs": 62.0,  
      "fat": 28.0,  
      "sugar": 7.0,  
      "sodium": 980.0  
    },  
    "servings": 4.0,  
    "ingredients": ["beef", "corn tortilla", "lime", "cilantro", "onion"]  
  },  
  {  
    "id": "R2",  
    "label": "Ropa Vieja",  
    "macros": {  
      "calories": 690.0,  
      "protein": 48.0,  
      "carbs": 76.0,  
      "fat": 22.0,  
      "sugar": 9.0,  
      "sodium": 940.0  
    },  
    "servings": 4.0,  
    "ingredients": ["beef", "rice", "tomato", "pepper", "onion"]  
  }  
]
```

```
},
{
  "id": "R31",
  "label": "Asian Beef & Lettuce Wraps",
  "macros": {
    "calories": 610.0,
    "protein": 46.0,
    "carbs": 48.0,
    "fat": 24.0,
    "sugar": 6.0,
    "sodium": 1137.0
  },
  "servings": 4.0,
  "ingredients": ["beef", "lettuce", "rice", "ginger", "tamari"]
},
{
  "id": "R35",
  "label": "Ropa Vieja",
  "macros": {
    "calories": 650.0,
    "protein": 49.0,
    "carbs": 70.0,
    "fat": 20.0,
    "sugar": 3.3,
    "sodium": 890.0
  },
  "servings": 4.0,
  "ingredients": ["beef", "rice", "tomato", "pepper", "onion"]
},
{
  "id": "R41",
  "label": "Berry Chia Breakfast Bowl",
  "macros": {
    "calories": 520.0,
    "protein": 24.0,
    "carbs": 72.0,
    "fat": 18.0,
    "sugar": 13.0,
    "sodium": 160.0
  },
}
```

```

    "servings": 2.0,
    "ingredients": ["chia", "berries", "banana", "almond milk", "oats"]
  },
  {
    "id": "R43",
    "label": "Sweet Potato Egg Hash",
    "macros": {
      "calories": 610.0,
      "protein": 32.0,
      "carbs": 68.0,
      "fat": 24.0,
      "sugar": 10.0,
      "sodium": 520.0
    },
    "servings": 4.0,
    "ingredients": ["egg", "sweet potato", "spinach", "olive oil", "pepper"]
  },
  {
    "id": "R44",
    "label": "Chicken Rice Bowl",
    "macros": {
      "calories": 680.0,
      "protein": 54.0,
      "carbs": 72.0,
      "fat": 18.0,
      "sugar": 7.0,
      "sodium": 710.0
    },
    "servings": 4.0,
    "ingredients": ["chicken", "rice", "pepper", "onion", "olive oil"]
  }
]

```

RULES:

1. Exactly 7 days; 3-5 meals/day; include breakfast/lunch/dinner daily.
2. Breakfast: choose recipes you judge breakfast-suitable from label, ingredients, preparation style, and context when available. Do not choose breakfast solely for macro density.
3. Medical constraints are mandatory.
4. Servings allowed: 0.5, 1.0, 1.5, 2.0.

5. Max 2 uses per recipe ID/week; count before answering.
6. Add snack/dessert only when useful.
7. Output recipe IDs only, not full recipe objects.

OUTPUT_SCHEMA:

```
{
  "goalStatus": "string",
  "summary": "string",
  "days": [
    {
      "day": 1,
      "meals": [
        {
          "slot": "breakfast|lunch|dinner|snack|dessert",
          "recipeId": "R1",
          "servings": 1.0,
          "reason": "short practical reason"
        }
      ],
      "dayRationale": "short rationale"
    }
  ]
}
```

CURRENT_PLAN:

```
[
  {
    "day": 2,
    "meals": [
      {
        "slot": "breakfast",
        "recipeId": "R43",
        "servings": 1.5
      },
      {
        "slot": "lunch",
        "recipeId": "R31",
        "servings": 2.0
      },
      {

```

```
    "slot": "dinner",
    "recipeId": "R2",
    "servings": 1.0
  },
  {
    "slot": "snack",
    "recipeId": "R41",
    "servings": 1.0
  }
],
"totals": {
  "calories": 3355.0,
  "protein": 212.0,
  "carbs": 348.0,
  "fat": 128.0,
  "sugar": 47.0,
  "sodium": 4249.0
}
},
{
  "day": 4,
  "meals": [
    {
      "slot": "breakfast",
      "recipeId": "R41",
      "servings": 2.0
    },
    {
      "slot": "lunch",
      "recipeId": "R44",
      "servings": 1.5
    },
    {
      "slot": "dinner",
      "recipeId": "R31",
      "servings": 1.0
    },
    {
      "slot": "snack",
      "recipeId": "R35",
```

```
    "servings": 1.0
  }
],
"totals": {
  "calories": 3280.0,
  "protein": 198.0,
  "carbs": 354.0,
  "fat": 116.0,
  "sugar": 42.3,
  "sodium": 3272.0
}
}
```

VALIDATION_ISSUES:

```
{
  "status": "REVISE",
  "issues": [
    {
      "type": "MEDICAL_CAP_EXCEEDED",
      "day": 2,
      "severity": "HIGH",
      "message": "Sugar exceeds the medical cap."
    },
    {
      "type": "MEDICAL_CAP_EXCEEDED",
      "day": 2,
      "severity": "HIGH",
      "message": "Carbs exceeds the medical cap."
    },
    {
      "type": "MEDICAL_CAP_EXCEEDED",
      "day": 4,
      "severity": "HIGH",
      "message": "Carbs exceeds the medical cap."
    },
    {
      "type": "RECIPE_REPETITION",
      "day": 0,
      "severity": "HIGH",
```

```
    "message": "Recipe R31 appears 3 times."
  }
]
}
```

NUMERIC_REPAIR_GUIDANCE:

```
{
  "passingDays": [1, 3, 5, 6, 7],
  "problemDays": [
    {
      "day": 2,
      "currentTotals": {
        "calories": 3355.0,
        "protein": 212.0,
        "carbs": 348.0,
        "fat": 128.0,
        "sugar": 47.0,
        "sodium": 4249.0
      },
      "validationIssues": [
        {
          "type": "MEDICAL_CAP_EXCEEDED",
          "day": 2,
          "severity": "HIGH",
          "message": "Sugar exceeds the medical cap."
        },
        {
          "type": "MEDICAL_CAP_EXCEEDED",
          "day": 2,
          "severity": "HIGH",
          "message": "Carbs exceeds the medical cap."
        }
      ],
      "performanceDeficits": [],
      "medicalCapExcesses": [
        {
          "nutrient": "Sugar",
          "current": 47.0,
          "maximum": 45.0,
          "excess": 2.0,
```

```
    "unit": "g"
  },
  {
    "nutrient": "Carbs",
    "current": 348.0,
    "maximum": 350.0,
    "excess": 0.0,
    "unit": "g"
  }
]
},
{
  "day": 4,
  "currentTotals": {
    "calories": 3280.0,
    "protein": 198.0,
    "carbs": 354.0,
    "fat": 116.0,
    "sugar": 42.3,
    "sodium": 3272.0
  },
  "validationIssues": [
    {
      "type": "MEDICAL_CAP_EXCEEDED",
      "day": 4,
      "severity": "HIGH",
      "message": "Carbs exceeds the medical cap."
    }
  ],
  "performanceDeficits": [],
  "medicalCapExcesses": [
    {
      "nutrient": "Carbs",
      "current": 354.0,
      "maximum": 350.0,
      "excess": 4.0,
      "unit": "g"
    }
  ]
}
```



```

],
"globalIssues": [
  {
    "type": "RECIPE_REPETITION",
    "day": 0,
    "severity": "HIGH",
    "message": "Recipe R31 appears 3 times."
  }
]
}

```

TASK:

Repair the plan with minimal changes. Priority: medical caps, breakfast/lunch/dinner present, breakfast appropriateness by your judgment, calorie/protein minimums, repetition/variety. Use NUMERIC_REPAIR_GUIDANCE exactly. Keep passing days unchanged unless needed for global repetition. Max 2 uses per recipe/week. Prefer 18-24 unique IDs when enough candidates exist. Return the complete corrected 7-day plan, not a patch.

- Одговор:

```

{
  "goalStatus": "HYPERTROPHY",
  "summary": "Repaired 7-day hypertrophy plan. The revised version reduces high-carbohydrate and high-sugar combinations on problem days, lowers repeated use of R31, and keeps breakfast/lunch/dinner structure intact.",
  "days": [
    {
      "day": 1,
      "meals": [
        {
          "slot": "breakfast",
          "recipeId": "R41",
          "servings": 1.0,
          "reason": "Breakfast-suitable chia bowl provides steady carbohydrates with low sodium."
        },
        {
          "slot": "lunch",

```

```

    "recipeld": "R44",
    "servings": 1.0,
    "reason": "Chicken Rice Bowl provides high protein and moderate
carbohydrates."
  },
  {
    "slot": "dinner",
    "recipeld": "R2",
    "servings": 1.0,
    "reason": "Ropa Vieja adds protein and rice-based energy for hypertrophy."
  },
  {
    "slot": "snack",
    "recipeld": "R35",
    "servings": 0.5,
    "reason": "Small serving adds calories without pushing sugar too high."
  }
],
"dayRationale": "Balanced day using low-sugar, high-protein options."
},
{
  "day": 2,
  "meals": [
    {
      "slot": "breakfast",
      "recipeld": "R43",
      "servings": 1.0,
      "reason": "Sweet Potato Egg Hash remains the best breakfast fit while reducing
total carbohydrates from the previous version."
    },
    {
      "slot": "lunch",
      "recipeld": "R44",
      "servings": 1.0,
      "reason": "Chicken Rice Bowl replaces the oversized R31 portion to lower sugar
and carbohydrates while preserving protein."
    },
    {
      "slot": "dinner",
      "recipeld": "R2",

```

```

      "servings": 1.0,
      "reason": "Moderate serving supports calories and protein without exceeding the
carbohydrate cap."
    },
    {
      "slot": "snack",
      "recipeId": "R35",
      "servings": 0.5,
      "reason": "Half serving adds extra calories with low sugar."
    }
  ],
  "dayRationale": "Repair reduces R31 usage and lowers sugar/carbs while keeping
the day protein-forward."
},
{
  "day": 3,
  "meals": [
    {
      "slot": "breakfast",
      "recipeId": "R41",
      "servings": 1.0,
      "reason": "Breakfast-suitable bowl adds controlled carbohydrates."
    },
    {
      "slot": "lunch",
      "recipeId": "R1",
      "servings": 1.0,
      "reason": "Steak tacos add high-quality protein and fats."
    },
    {
      "slot": "dinner",
      "recipeId": "R7",
      "servings": 1.0,
      "reason": "Turkey Bean Chili provides protein and fiber."
    }
  ],
  "dayRationale": "Keeps protein high while avoiding excessive recipe repetition."
},
{
  "day": 4,

```

```

"meals": [
  {
    "slot": "breakfast",
    "recipeId": "R43",
    "servings": 1.0,
    "reason": "Egg and sweet potato breakfast provides better sugar control than a
double serving of R41."
  },
  {
    "slot": "lunch",
    "recipeId": "R44",
    "servings": 1.0,
    "reason": "Reduced serving lowers carbs while preserving protein."
  },
  {
    "slot": "dinner",
    "recipeId": "R35",
    "servings": 1.0,
    "reason": "Lower-sugar Ropa Vieja supports dinner calories without exceeding
the carb cap."
  },
  {
    "slot": "snack",
    "recipeId": "R1",
    "servings": 0.5,
    "reason": "Small steak taco serving adds protein and calories with controlled
carbs."
  }
],
"dayRationale": "Repair lowers carbohydrates below the cap by reducing R41 and
avoiding R31."
},
{
  "day": 5,
  "meals": [
    {
      "slot": "breakfast",
      "recipeId": "R41",
      "servings": 1.0,
      "reason": "Breakfast-appropriate and low sodium."
    }
  ]
}

```

```

    },
    {
      "slot": "lunch",
      "recipeId": "R2",
      "servings": 1.0,
      "reason": "Balanced beef and rice lunch."
    },
    {
      "slot": "dinner",
      "recipeId": "R7",
      "servings": 1.0,
      "reason": "Turkey Bean Chili gives protein and fiber."
    },
    {
      "slot": "snack",
      "recipeId": "R35",
      "servings": 0.5,
      "reason": "Low-sugar calorie support."
    }
  ],
  "dayRationale": "Stable protein and calories with moderate carbohydrate exposure."
},
{
  "day": 6,
  "meals": [
    {
      "slot": "breakfast",
      "recipeId": "R43",
      "servings": 1.0,
      "reason": "Breakfast-suitable egg and sweet potato meal."
    },
    {
      "slot": "lunch",
      "recipeId": "R31",
      "servings": 0.5,
      "reason": "Reduced R31 portion keeps weekly repetition and carbohydrate load
controlled."
    },
    {
      "slot": "dinner",

```

```

      "recipeld": "R4",
      "servings": 1.0,
      "reason": "Chicken and rice supports muscle recovery."
    },
    {
      "slot": "snack",
      "recipeld": "R1",
      "servings": 0.5,
      "reason": "Adds protein and fats without excessive carbohydrates."
    }
  ],
  "dayRationale": "Uses R31 only once at a reduced portion and balances with leaner meals."
},
{
  "day": 7,
  "meals": [
    {
      "slot": "breakfast",
      "recipeld": "R41",
      "servings": 1.0,
      "reason": "Breakfast-suitable chia bowl helps maintain energy."
    },
    {
      "slot": "lunch",
      "recipeld": "R44",
      "servings": 1.0,
      "reason": "Chicken Rice Bowl provides protein and carbohydrates."
    },
    {
      "slot": "dinner",
      "recipeld": "R2",
      "servings": 1.0,
      "reason": "Ropa Vieja closes the week with a balanced high-protein dinner."
    }
  ],
  "dayRationale": "Concludes with controlled sugar and carbohydrate intake while keeping protein adequate."
}
]

```

}

- Доколку планот е поправен или пак доколку лимитот на итерации за поправка на планот е достигнат, се испраќа планот до корисникот.

3. Краток заклучок:

Поради ограничени ресурси и грешки во мапирањето на рецептите во базата, може да се заклучи дека директниот повик до Gemini резултираше со несоодветни планови. Исто така може да се примети дека честопати рецептите за појадок не се соодветни. Ова пробавме да го решиме преку посебен прашалник до база каде ќе се пребаруваат рецепти кои содржат имиња на состојки соодветни за појадок, како и преку построги наредби во prompt-от на агентите, но сепак не добивме несоодветни предлози.